

2023

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer from **all** the section as directed.*

Section-A

Multiple choices questions

(Compulsory)

1. Choose the correct answer from the following
multiple choice questions : $2 \times 10 = 20$

(i) A linked list type that navigates for an item in forward and backward manner is called ?

(a) Singly linked list

☒ (b) Doubly linked list

(c) Circular linked list

(d) None of the above

(ii) Process of deletion an element from a stack is called _____

(a) Create

(b) Exit

☒ (c) Pop

(d) Push

(iii) In a stack, if a user tries to remove an element from empty stack is called _____.

(a) Garbage collection

(b) Empty condition

(c) Overflow

(d) Underflow

(iv) Pushing an element into a stack which is full triggers _____ condition.

(a) Crash

(b) Underflow

(c) Overflow

(d) Userflow

(v) Which data structure is used to implement recursion ?

(a) Queue

(b) Stack

(c) Linked list

(d) Array

(vi) Sparse matrix is array with _____

(a) Maximum non-zero elements

☒ (b) Maximum zero elements

(c) Minimum zero elements

(d) None of the above

(vii) Which of the data structure is linear ?

(a) Linked list

(b) Stack

(c) Queue

☒ (d) All of the above

(viii) What will be the prefix of the following ?

Infix : DGBAHEICF

Prefix : ABDGCEHIF

(a) DBGHIEFCA

☒ (b) GDBHIEFCA

(c) GDBHEIFCA

(d) GDBECHIFA

(ix) What is the initial value of top variable ?

(a) 0

☒ (b) -1

(c) 1

(d) Size

(x) How many control variable are there in a queue ?

(a) 1

☒ (b) 2

(c) 3

(d) 4

Section-B

Short answer type questions

Answer any **four** questions of the following :

$$5 \times 4 = 20$$

- ☒ 2. Differentiate between stack and queue in array implementation.

- ✓ 3. WAP for Binary Search Operation of a sorted array.
- ✓ 4. WAP for Bubble sort.
- ✓ 5. WAP to find out the sum of first 10 natural numbers using recursive function.
- ✓ 6. WAP for multiplication of two 3×3 matrices.
- ✓ 7. Explain the array representation of sparse matrix.
- ✓ 8. WAF for insertion and deletion of an element in stack using pointer implementation.

Section-C

Long answer type questions

Answer any **two** questions of the following :

$$15 \times 2 = 30$$

- ✓ 9. What is BST ? Why AVL tree is used ? What are the four rotational techniques in AVL tree ?

✓10. WAP for Doubly Linked List implementation for insertion, Deletion, search, count and display operation using C.

✓11. WAP for array implementation for stack for push, pop, search, count, display operations using C.

12. Write short notes on any **three** of the following :

✓(a) Hashing

(b) Threaded Binary Tree

(c) Priority Queue

(d) Skip List

✓(e) Sparse Matrix

✓(f) BST

—————x—————